

EURODATA

Interactive European Social & Economic Data Explorer

COM-480 Data Visualization
Process Book

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Project Goal

Understanding European Diversity Through Religious Data

The origins of religion can be traced back to at least 100,000 years ago, and it has continuously and profoundly influenced the development and continuation of human civilization. According to incomplete statistics, there are currently approximately 4,000 different religions and belief systems worldwide, closely linked to human social life, cultural traditions, and economic activities. Therefore, this project aims to use data visualization to display the distribution, evolution, and societal impact of religious patterns. We focus on linking these religious patterns with the various social questions, and we specifically showcase this with European data.

Dataset

Pew Research Center & European Social Survey Data

Primary Data Sources

All datasets used in this project are sourced from the **Pew Research Center**. Pew Research Center is a nonpartisan fact tank that informs the public about the issues, attitudes and trends shaping the world. This project primarily selects data from the International religion survey data. We used two European datasets - Western Europe survey and Central and Eastern Europe survey.

Western Europe Survey

Period: April to August 2017

Sample: 26,000+ telephone interviews

Coverage: 16 countries, 14 languages

Focus: Role of religion in public and private life, views toward religious minorities

Central & Eastern Europe Survey

Period: June 2015 to July 2016

Sample: 31,000+ face-to-face interviews

Coverage: 19 countries, 17 languages

Focus: Religion in public life and national identity post-Soviet Union

Dataset Challenges

These datasets are very broad and informative, but they were not unified, which restricted us in what we could show in our visualizations. If one dataset had the evaluation of respondent's happiness level, the other did not, and that meant that this kind of data could not be included.

Temporal Data - European Social Survey (ESS)

We also found datasets created by the European Social Survey (ESS) to showcase the temporal aspect of religions. We had 11 different datasets from 2002 to 2024 conducted every two years. We chose to use these datasets because the social climate changes and religious populations, especially with this information age evolve very quickly.

Exploratory Data Analysis

Understanding Dataset Boundaries and Potential

In Milestone 1, we tried to find the boundaries of the datasets and find what exactly this kind of survey data can show to us. A lot of survey questions were specific to each country, such as how Lithuanians perceive the religious minorities population at the time of the question compared to the time in the Soviet Union. We discovered that these questions were not valuable as they did not give an insight into how the countries compare.

Sample Data Analysis

Region	Christian	Muslim	Atheist	Other
Eastern Europe	78%	9.1%	3.5%	9.4%
Western Europe	68.1%	1%	10.7%	20.2%

Design Choices

Website Layout & Visual Identity

Website Layout

From the start of the project we had an idea to compare countries, thus our main focus was always the map. To display the comparisons, we chose to color the countries according to their most common answers such that the user could immediately see results on a surface level. The goal was to make the experience of finding and interacting with data as smooth and easy as possible, thus we opted for a scrollable layout.

Website Color Scheme

#667eea
Primary

#764ba2
Secondary

#2E86AB
Accent 1

#A23B72
Accent 2

#F18F01
Highlight

For our color scheme we chose purple gradient with blue. As religion and various opinions are very abstract topics we wanted something neutral but engaging. Therefore we opted for a smooth gradient coloring that we kept consistent with the visualizations.

Storytelling

User Journey Through European Diversity

We wanted to tell the story of how religions impact the way people perceive their countries and what beliefs they hold. However, during the project and our storytelling, we found that religions do not divide people as much as we thought they were. Therefore, we wanted the user to experience the same realization as we had through various topics such as economic state perception and abortion rights in the countries.

User Journey

1. Introduction & Expectations

First, we start with the introduction to the user what visualizations they can expect by listing all types of data the website showcases. We also add some leading questions which creates expectation to find these answers later on and acts as a hook.

2. Mission & Context

Second, we state our mission to show the diversity of European countries, and availability of endless comparisons. In addition to that, we explain the categories mentioned in the introduction more in depth, so that the user could understand what they will see in the map. Then, we explain the datasets where the data was extracted, and how it was processed.

3. Interactive Exploration

Lastly, we let the user explore the map by hovering over the map when their desired comparison category is selected, and discover the data more in depth. If the user desires, they can compare two countries over the selected category to have more visual understanding where the difference lies.

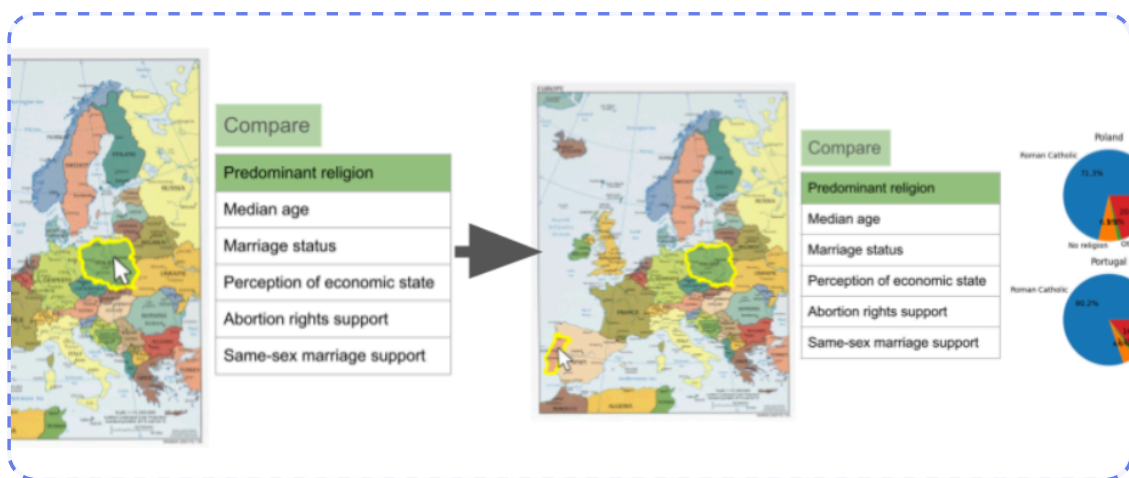
Visualizations

Interactive Data Exploration Tools

We initially started with the goal to implement one interactive map that would change the coloring depending on the selected topic. In addition to that, we wanted users to interact with the map and discover deeper information by comparing selected countries.

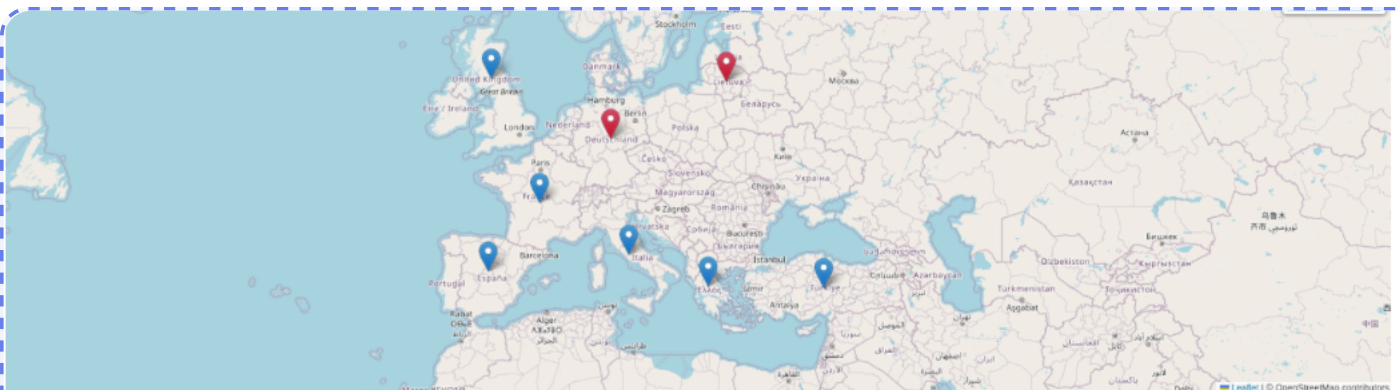
Sketches

Our initial sketches were barebones with the ideas for functionality and to be convincing. Our design was not fluid or intuitive to use, but we wanted our ideas to be represented in the sketches.



Once we made our prototype for Milestone 2 without using D3, we developed ideas how to guide the user's eye without the additional test explanations and tutorials. We needed to make our website intuitive.

Implementation



Germany

Lithuania

Protestant Christianity

Catholic Christianity

Judaism

Catholic Christianity

Atheist

Orthodox Christianity

Atheist

Detailed Comparison

Religious Denominations Over Time

Since our map is based on only a one year survey, we wanted the user to see how the religious belonging and denominations evolved in detail over the past 20 years. Therefore, we created a bar chart that compares all countries from that year's ESS data, and shows the detailed distribution of religious denominations. This is an interactive bar chart, which the user can hover and see the data without having to look at the axes carefully. For easier reading, the data is sorted from the highest values to lowest, which means that for religious belonging the first country shown is always going to be with the highest religious population percentile, and for the religious denomination will be the country with a single stand-out denomination.

In addition to the bar chart, we added another map, which does not include the opinions of surveyees, but only their religious associations over the years. After gaining insight how the religious beliefs could have impacted surveyed opinions in the Pew Research data the user can gain insight how the same countries trended over the years.

Challenges

Technical and Design Obstacles



Dataset Selection

Our initial challenge once we decided to explore religions was picking the right dataset. There are a lot of census datasets from different countries, but the census usually does not represent the population's opinions and views. In addition to that, the census is not standardized across different countries. Therefore we landed on Pew Research Center datasets from 2016 and 2017 as it was open source and covered many topics.

Data Integration

Next challenge was finding the data that is shared across both datasets, and extracting it, as data was the aggregated answers to the surveys. Once the data was selected, it had to be filtered, cleaned and processed for visualizations.

Geographic Mapping

When it came to visualizing our data, we found that some countries that were drawn on the map and were in the processed data, were not showing any results. This was because of misalignment of country names between our datasets and topological data that we decided to use.

Bias and Data Selection

One of the more implicit challenges we had to solve was choosing what data we want to represent from the surveys. We all have our biases and interests, but we ended up choosing data that we thought would be impacted by the religious beliefs such as abortion rights, same-sex marriage, and marriage status of the respondents.

Temporal Data Processing

Another challenge we faced was that in the temporal datasets, the countries in one year were not guaranteed to be in the other year. This missing data meant we had a lot of empty data gaps to handle, which we ended up visualizing as missing data in the map, and rerendering the bar chart once the year is selected.

Peer Assessment

Team Member Contributions

Alaa El Ouahabi

Data collection for temporal data (religions belonging + nomination) + cleaning it

Website design

Code the whole website

Work on all the visualisations

Code the html for the final report

Record the screencast

Martynas Rimkevičius

Initial dataset analysis + feasibility assessment

Survey data processing and merging for the visualization use

Initial website skeleton (layout + visualization ideas)

Website hosting configuration

Map interactivity implementation (showing tooltip) and loading data into the map

Wrote the process book and made visuals of the evolution

Zheyang Lyu

Join team discussions

Made some visualisation code but it's not perfect

Read code and report